

Spatial localization and quantitative determination of lignin in chickpea root cell wall in response to disease

Materials

1. Agarose solution: 4%–5% (w/v) solution in a small tray for embedding.
2. Cryo-microtome: Leica Biosystems, CM1800.
3. Vibratome: Leica Biosystems VT1000S.
4. Phloroglucinol stain: 2% phloroglucinol solution in 80 mL of 20% ethanol and 20 mL of concentrated hydrochloric acid.
5. Basic fuchsin stain: Prepared according to Khandal et al. (2020).
6. Microscopes: Nikon Eclipse 80i fluorescent microscope and Leica Biosystems SP8 confocal fluorescence microscope.
7. Extraction buffers: Solution I (50 mM Potassium phosphate buffer, pH 7.0), Solution II (1% v/v Triton X-100 in Solution I), and Solution III (1M NaCl in Solution I).
8. Solvents and chemicals: Liquid nitrogen, distilled water, acetone, glacial acetic acid, 25% (v/v) acetyl bromide in glacial acetic acid, 2M NaOH, and 5M hydroxylamine-HCl.
9. Equipment: Dry bath, 60°C oven, vacuum desiccator, centrifuge, and UV-Visible spectrophotometer.

Tissue sectioning

1. Prepare transverse sections of the root tissue using a cryo-microtome.
2. For longitudinal sections, fix the root tissue samples with 4%–5% agarose solution in a small tray.
3. Trim the embedded tissue block into a cube and secure it onto the tissue holder of a vibratome.

4. Cut sections to a thickness of approximately 50 μm using a blade speed of 1.5 mm/s and a frequency of 60 Hz.

Histochemical staining and microscopy

1. For phloroglucinol staining, immerse the chickpea sections in the 2% phloroglucinol solution for 3–5 minutes.
2. Wash the sections once with distilled water for 30 seconds.
3. Observe the phloroglucinol-stained sections under a fluorescent microscope.
4. For basic fuchsin staining, perform the protocol following the method described in Khandal et al. (2020).
5. Observe the basic fuchsin-stained sections under a confocal fluorescence microscope.

Cell wall extraction and washing

1. Crush frozen root samples into a powder using liquid nitrogen.
2. Homogenize the crushed tissue with Solution I.
3. Centrifuge the mixture at 1400 g for 5 minutes and discard the supernatant.
4. Wash the remaining pellet two times with Solution I, centrifuging at 1400 g for 5 minutes after each wash.
5. Wash the pellet three times with Solution II.
6. Wash the pellet two times with Solution III.
7. Wash the pellet two times with distilled water.
8. Wash the pellet two times with acetone.
9. Dry the washed pellet in an oven at 60°C for 24 hours.
10. Cool the dried pellet completely under vacuum in a desiccator to obtain the protein-free cell wall extract.

Lignin solubilization and quantification

1. Weigh out 20 mg of the dried protein-free cell wall extract and incubate it with 0.5 mL of 25% acetyl bromide solution in a 70°C dry bath for 30 minutes.
2. Transfer the sample immediately onto ice to cool.
3. Add 0.9 mL of 2 M NaOH, 0.1 mL of 5 M hydroxylamine-HCl, and 5 mL of glacial acetic acid to the mixture to completely solubilize the lignin.
4. Centrifuge the final mixture at 1400 g for 5 minutes.
5. Collect the supernatant and read the absorbance at 280 nm using a UV spectrophotometer.
6. Calculate the total lignin content using an absorptivity (ϵ) value of 22.9 g⁻¹ L cm⁻¹ (Moreira-Vilar et al. 2014) and express the results as mg of lignin g⁻¹ cell wall extract.

References

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